Schema and data dictionary

Schema and data dictionary I

The descriptor's `fields` list is the dataset's **data dictionary**: it names each column of the primary measurement table and declares its type, nullability, optional unit, and value constraints. fig. 2 renders this dictionary directly from the descriptor via `schema_table_rows()` in `src/data_descriptor/figures.py`.

Schema and data dictionary I

Field schema / data dictionary

field	type	nullable	unit	constraint
sample_id	string	no	—	pattern ^S[0-9]{3}$
subject_id	string	no	—	pattern ^P[0-9]{3}$
group	category	no	—	{control, treatment}
value	number	no	normalized_score	[0, 1]
collected_on	date	no	—	—
instrument	category	no	—	{sensor_a, sensor_b}

Figure 2: Field schema / data dictionary: one row per declared field, with type, nullability, unit, and a compact constraint label, from `schema_table_rows()`. Identifier fields carry regular-expression patterns; the categorical fields carry closed enumerations; the quantitative field carries a unit and numeric bounds. The table is generated from `data/example_descriptor.json` and cannot silently disagree with it.

Field contract I

The six declared fields exercise the constraint vocabulary the validator understands:

Field contract I

- **sample_id** (string, not null) — the primary key, constrained to the pattern `^S[0-9]{3}$`.
- **subject_id** (string, not null) — foreign key into `fixtures/subjects.csv`, constrained to `^P[0-9]{3}$`.
- **group** (category, not null) — a closed enumeration of `control` and `treatment`.
- **value** (number, not null) — the measurement, carrying the unit `normalized_score` and numeric bounds `[0, 1]`.
- **collected_on** (date, not null) — an ISO-8601 collection date.
- **instrument** (category, not null) — a closed enumeration of `sensor_a` and `sensor_b`.

Schema fingerprint I

`descriptor_fingerprint()` reduces the field list to a stable, order-independent sha256 fingerprint over `(name, type, nullable)` triples. Reordering the fields leaves the fingerprint unchanged, so two descriptors with the same schema hash identically regardless of authoring order — the property is asserted directly in the test suite. The fingerprint is what lets a release manifest reference "the schema" by a single value that changes if and only if the field contract changes.

Constraint validation I

The validator flags quantitative fields that lack a unit, categorical fields that lack allowed values, enumerations that are present but empty, numeric bounds where the minimum exceeds the maximum, and patterns declared on non-text fields. On the shipped descriptor none of these fire — the data dictionary above is complete, so the readiness score is unpenalised (see sec. 6).